

European Quantum Trusted Data Sharing Enterprise Blockchain Infrastructures

In the European market, the integration of tokenless blockchain middleware with quantum-safe solutions is primarily driven by national telecommunications giants (ISPs) collaborating with European cybersecurity leaders and by EU-funded initiatives such as EuroQCI.

Key European Internet Services Providers (ISPs) and the specific infrastructures they are using to provide quantum-safe, trusted data sharing for enterprises and SMEs:

1. Orange Business (France) – "Quantum Defender"

Orange has taken a leading role in commercializing quantum-safe connectivity for the European enterprise market.

- **Blockchain Integration:** Orange utilizes blockchain for **managed security and traceability** (specifically through its Cyberdefense arm) to ensure the integrity of security logs and transaction audits.
- **Quantum Solution:** Their **Quantum Defender** service uses a hybrid approach:
 - **Post-Quantum Cryptography (PQC):** Software-based encryption (ML-KEM/Kyber) to protect standard SD-WAN and Ethernet traffic.
 - **Quantum Key Distribution (QKD):** Partnered with **Toshiba Europe** to provide physical quantum keys for critical links.
- **SME Benefit:** It allows SMEs to "future-proof" their data today, protecting against "Harvest Now, Decrypt Later" (HNDL) attacks.

2. Deutsche Telekom & Telefónica (Germany/Spain) – QSAFE & EuroQCI

These two operators are the backbone of the **EuroQCI (European Quantum Communication Infrastructure)**, which aims to build a quantum-secure "shield" across all 27 EU member states.

- **Blockchain Integration:** As governing members of **Hedera** and key contributors to **EBSI (European Blockchain Services Infrastructure)**, they use a **tokenless middleware** layer for "Verifiable Credentials." This allows SMEs to share authenticated business records (such as tax data or supply chain origin) without relying on a centralized database.
- **Quantum Solution:** They are currently migrating these blockchain nodes to **PQC-based signatures** (such as Dilithium) to ensure that the identity records stored on the ledger cannot be forged by future quantum computers.
- **European Partners:** They collaborate closely with **Thales** (France) and the **AIT Austrian Institute of Technology**.

3. Telia & Elisa (Nordics/Baltics) – Guardtime KSI Integration

In Northern Europe and Estonia, ISPs often leverage **Guardtime**, a European leader in quantum-immune blockchain technology.

- **Blockchain Integration:** **Guardtime's KSI (Keyless Signature Infrastructure)** is a

unique, **tokenless, hash-based blockchain**. Because it relies solely on hash functions rather than asymmetric encryption (RSA/ECC), it is mathematically **quantum-immune**.

- **Quantum Solution:** Telia and local Baltic ISPs integrate KSI as a middleware for SMEs in the energy and government sectors. It acts as a "Trusted Data Sharing" layer where every piece of data is "signed" by the blockchain.
- **SME Benefit:** It is purely infrastructure-focused (no tokens or coins), providing a simple API for SMEs to prove the integrity of their data in a post-quantum world.

4. Quantum Blockchains (Poland/EU) – POSEIDON Project

This is a newer European contender specifically designed to bridge the gap between ISPs and quantum safety.

- **Blockchain Integration:** Their **QSB (Quantum Safe Blockchain)** is a tokenless middleware designed for European Digital Identity (EUDI) projects.
- **Quantum Solution:** It is part of the **POSEIDON EU project**, which integrates PQC and QKD directly into the blockchain consensus layer.
- **ISP Connection:** They partner with local infrastructure providers to secure **Industry 4.0** environments (like autonomous robotics in Poland) where data integrity must survive the transition to the quantum era.

Summary of European Alliances

ISP / Provider	Blockchain Middleware	Quantum Safety Partner	Target Sector
Orange Business	Orange Cyberdefense	Toshiba Europe / Thales	Finance & Manufacturing
Deutsche Telekom	EBSI / Hedera	Thales / AIT	Gov & Public Services
Telia / Elisa	Guardtime KSI	(Inherent Hash-based)	Energy & e-Government
Telefónica	TrustOS (Middleware)	QSAFE Consortium	IoT & Supply Chain
Proton (Swiss)	PGP-based Middleware	Internal PQC R&D	Privacy-first SMEs

Guardtime, Thales, and EBSI-aligned ISPs, are designed specifically for **EU Digital Sovereignty**. They ensure that the encryption keys and the data-sharing audit trails remain within European legal jurisdiction, adhering strictly to **NIS2** and the **Cyber Resilience Act (CRA)**.

As the quantum computing era approaches in 2026, the transition for SMEs (Small and Medium-sized Enterprises) centers on adopting **Interoperable, Private, and Secure (IPS)** blockchain middleware to mitigate the "cryptographic cliff" through **Post-Quantum Cryptography (PQC)** and **Quantum Key Distribution (QKD)**.

The following sources provide the technical, regulatory, and industrial foundation for these services, with a specific focus on tokenless infrastructure and European sovereign solutions.

Foundational & Regulatory Frameworks

These sources establish the necessity and timelines for the quantum transition within the UK and European Union, highlighting the "Harvest Now, Decrypt Later" (HNDL) threat.

- **National Cyber Security Centre (NCSC) & ENISA (2026):** These agencies have set clear migration milestones: discovery by 2028 and completion by 2035, with the EU calling for critical infrastructure transition to begin by late 2026 (Talan, 2026). They recommend specific NIST-approved algorithms, such as **ML-KEM-768** and **ML-DSA-65** (Talan, 2026).
- **Europol & FS-ISAC (2026):** Their joint report, *Prioritizing Post-Quantum Cryptography Migration Activities in Financial Services*, provides the SOTA prioritization framework for determining migration urgency based on quantum risk versus the time required for implementation (PQShield, 2026).

Key Quantum-Safe Architectures (IPS & Middleware)

These papers and industry releases describe the specific systems (e.g., Q-AmChain, KSI) that act as "plug-and-play" middleware for SMEs.

- **Ameritec IPS Q-AmChain & QB-CURE (2026):** A foundational commercial deployment expected in June 2026. Q-AmChain is a rebuilt **Proof-of-Authority (PoA)** blockchain that integrates PQC across every layer, including transaction validation and validator nodes (Ameritec IPS, 2026a). The **QB-CURE** system pairs these signatures with **biometric facial authentication** to fix the "single point of failure" in recovery phrases (Ameritec IPS, 2026b).
- **Guardtime KSI® Blockchain (2026):** A "foundational" European technology for tokenless data sharing. Unlike typical PQC, KSI is a **hash-based** infrastructure. Because it relies on one-way hash functions rather than asymmetric encryption, it is considered **mathematically quantum-immune** (Guardtime, 2026).

Recent State-of-the-Art (SOTA) Research

These 2025–2026 papers explore the technical feasibility of integrating quantum safety into industrial and terrestrial networks.

- **National Terrestrial QKD Planning (2026):** This research addresses how to plan national **Quantum Key Distribution (QKD)** networks within the **EuroQCI** framework, providing the first-order dimensioning estimates for the backbone of European quantum-safe communication (arXiv:2604.06764v1).
- **Blockchain & Quantum Machine Learning (2025):** A scoping review analyzing 266 studies that identifies the growing adoption of blockchain for **secure data logging** and the use of QML for anomaly detection, while highlighting interoperability as the primary

barrier for SME adoption (Adere et al., 2025).

Surveys & Strategy Reports

- **World Economic Forum (2026):** Focuses on "**Cryptographic Agility**"—the ability to swap algorithms without full architectural redesigns—as a strategic asset for business resilience during the migration (World Economic Forum, 2026).
 - **PQCrypto 2026 Conference:** The 17th International Conference on Post-Quantum Cryptography (held in April 2026 in Saint-Malo, France) serves as the definitive gathering for the latest academic SOTA on hash-based, lattice-based, and code-based primitives (PQCrypto, 2026).
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Relevant References

Ameritec IPS. (2026a, March 24). Ameritec IPS Unveils Q-AmChain, a Quantum-Resistant Blockchain Platform Ahead of June 2026 Deployment. *GlobeNewswire*.
<https://www.globenewswire.com/news-release/2026/03/24/3261020/0/en/ameritec-ips-unveils-q-amchain-a-quantum-resistant-blockchain-platform-ahead-of-june-2026-deployment.html>

Ameritec IPS. (2026b, February 23). Ameritec IPS Announces QAmChain Post-Quantum Blockchain and QB-CURE Biometric Wallet. *GlobeNewswire*.
<https://www.globenewswire.com/news-release/2026/02/23/3242307/0/en/Ameritec-IPS-Announces-QAmChain-Post-Quantum-Blockchain-and-QB-CURE-Biometric-Wallet-Upgrading-the-HEW-E-Ecosystem-for-the-Quantum-Era.html>

Adere et al. (2025). Blockchain and Quantum Machine Learning Approach for Securing Smart Water Management Systems: A Scoping Review. *SHIFRA* 2025, 141–202.
<https://www.researchgate.net/publication/398580394>

arXiv. (2026). *Towards National Quantum Communication in Europe: Planning and Sizing Terrestrial QKD Networks*. <https://arxiv.org/html/2604.06764v1>

Guardtime. (2026). *KSI Blockchain Technology: Quantum Immunity and Data Privacy*.
<https://guardtime.com/ksi>

PQShield. (2026, January 22). *Europol: Prioritizing PQC Migration in Financial Services*.
<https://pqshield.com/europol-prioritising-pqc-migration-in-financial-services/>

Talan. (2026, March 26). *Post-quantum cryptography in 2026*.
<https://www.talan.com/uk/en/post-quantum-cryptography-2026>

World Economic Forum. (2026, January 26). *Quantum-safe migration: An opportunity to modernize cryptography*.
<https://www.weforum.org/stories/2026/01/quantum-safe-migration-cryptography-cybersecurity/>